



Tailscale Remote Access Implementation

Overview

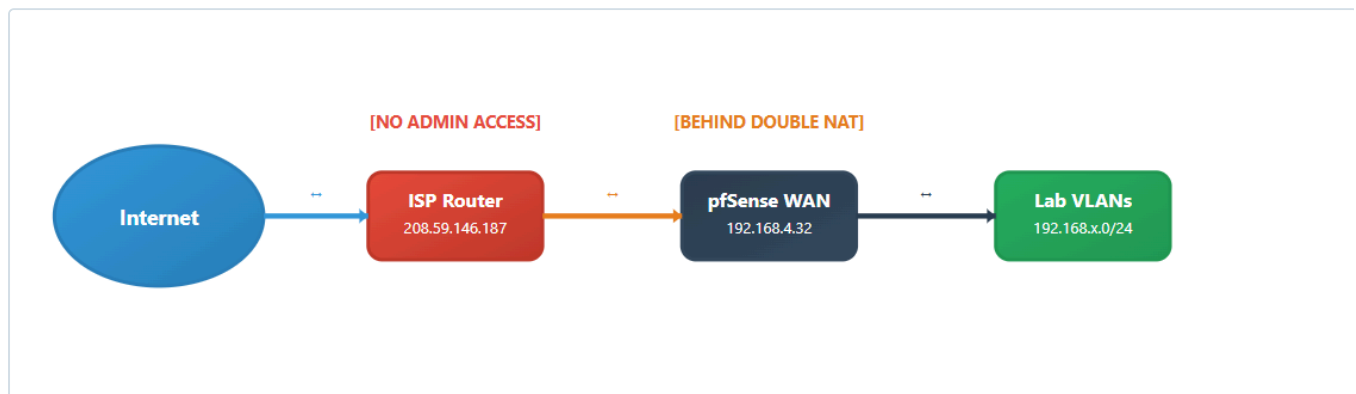
This document details the implementation of **Tailscale mesh VPN** as the remote access solution for the pfSense-based homelab. The implementation provides secure, zero-configuration remote access to all VLAN segments without requiring ISP cooperation or port forwarding, effectively bypassing the limitations of the ISP-controlled upstream router.

The Challenge: ISP Router Limitations

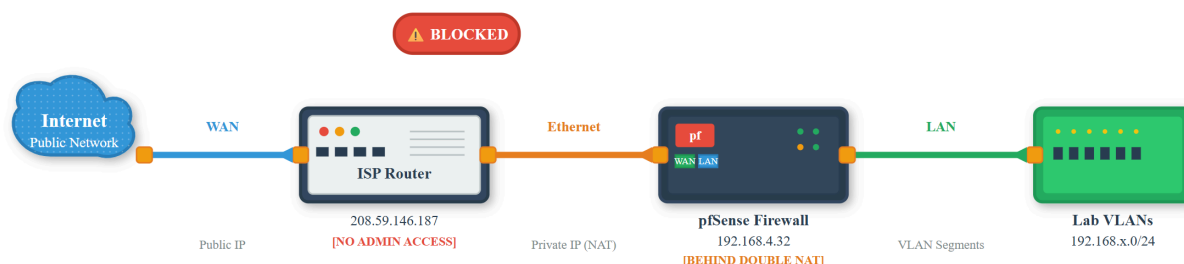
Network Topology Constraint

The lab's network architecture presented a critical challenge:

Internet ↔ ISP Router (208.59.146.187) [NO ADMIN ACCESS] ↔ pfSense WAN (192.168.4.32) [BEHIND DOUBLE NAT] ↔ Lab VLANs (192.168.x.0/24)



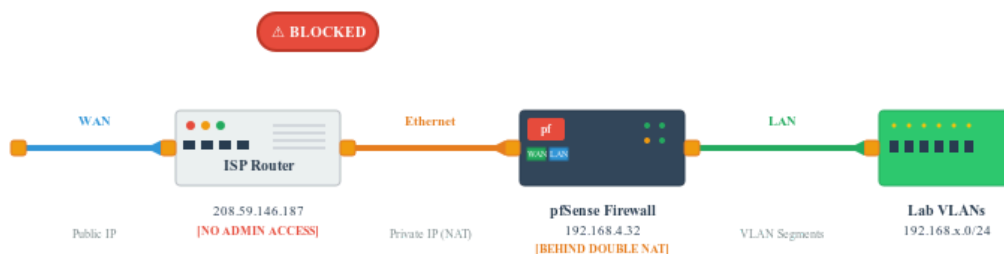
Network Topology Constraint - ISP Router Limitations



Network Path: Internet → ISP Router (No Access) → pfSense (Behind NAT) → Lab VLANs

Challenge: Traditional VPN solutions require port forwarding on ISP router (208.59.146.187) which is inaccessible

Network Topology Constraint - ISP Router Limitations



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Challenge: Traditional VPN solutions require port forwarding on ISP router (208.59.146.187) which is inaccessible

Specific Issues: - **No Administrative Access** to ISP-provided router - **pfSense behind NAT** at 192.168.4.32 instead of public IP - **Port forwarding impossible** for traditional VPN solutions - **Double NAT configuration** complicating connectivity

Solution Evaluation

Solution	ISP Access Required	VLAN Support	Decision
OpenVPN	✓ Yes (Port forwarding)	✓ Full	✗ Not viable
ZeroTier	✗ No	✗ Limited (1 route free)	✗ Insufficient
Tailscale	✗ No	✓ Unlimited	✓ Selected

Implementation Process

Phase 1: Tailscale Account Setup

Step 1: Account Creation 1. Registered at tailscale.com 2. Authenticated with a Google account as the identity provider 3. Completed user survey (Personal/Security use case) 4. Obtained free plan (100 devices, unlimited routes)

Step 2: Initial Dashboard Access - Accessed admin console at `login.tailscale.com/admin` - Confirmed the tailnet was created under that account - Prepared for device enrollment

Phase 2: pfSense Integration

Step 3: Package Installation

Connected to pfSense via SSH (`ssh admin@192.168.10.1`) and installed Tailscale:

```
# Update FreeBSD package repository
pkg update

# Install Tailscale package
pkg install -y tailscale

# Enable for automatic startup
sysrc tailscaled_enable="YES"
```

Step 4: Service Initialization Challenge

Initial service start failed due to FreeBSD configuration requirements:

```
# This failed initially
service tailscaled start
# Error: Cannot 'start' tailscaled. Set tailscaled_enable to YES in /etc/rc.conf
```

Solution - Manual Service Start:

```
# Create required directories
mkdir -p /var/lib/tailscale

# Start tailscaled manually
/usr/local/bin/tailscaled --state=/var/lib/tailscale/tailscaled.state --socket=/var/run/tailscale/

# Verify service is running
ps aux | grep tailscaled
```

Output confirmed successful start:

```
Program starting: v1.80.0, Go 1.23.3: ["/usr/local/bin/tailscaled", "--state=/var/lib/tailscale/ta
LogID: dac29c9f33762dad26aa0a7353807ca70e472d7e5d1fbabdb11ada7bbde44e3c
```

Step 5: Network Registration

Joined the Tailscale network and advertised all VLAN routes:

```
/usr/local/bin/tailscale up \  
--advertise-routes=192.168.10.0/24,192.168.20.0/24,192.168.30.0/24,192.168.40.0/24,192.168.50.0/  
--accept-routes \  
--hostname=pfsense-homelab
```

Authentication Process: - Command generated authentication URL - Opened URL in browser and completed OAuth authentication - pfSense appeared in admin console as "pfsense-homelab" - Assigned Tailscale IP: 100.94.120.6

Phase 3: Route Configuration

Step 6: Subnet Route Activation

In Tailscale admin console (login.tailscale.com/admin):

1. **Located pfSense device** in Machines list
2. **Clicked "Subnets" button** next to pfsense-homelab
3. **Found all routes pre-approved:** - ☒ 192.168.10.0/24 (Management VLAN) - ☒ 192.168.20.0/24 (BlueTeam VLAN) - ☒ 192.168.30.0/24 (RedTeam VLAN) - ☒ 192.168.40.0/24 (DevOps VLAN) - ☒ 192.168.50.0/24 (EnterpriseLAN VLAN) - ☒ 192.168.60.0/24 (Monitoring VLAN)

Note: Routes were automatically approved - no manual enablement required.

Phase 4: Client Device Setup

Step 7: Windows Client Installation

1. **Downloaded Tailscale** from official website
2. **Installed with default settings** on Windows machine
3. **Authenticated** using same Google account
4. **Device automatically appeared** as "nantwi" in admin console
5. **Assigned Tailscale IP:** 100.90.17.43

☒ Verification and Testing

Connectivity Tests

Test 1: VLAN Gateway Reachability

```
C:\> ping 192.168.10.1 # Management - SUCCESS  
C:\> ping 192.168.20.1 # BlueTeam - SUCCESS  
C:\> ping 192.168.30.1 # RedTeam - SUCCESS  
C:\> ping 192.168.40.1 # DevOps - SUCCESS  
C:\> ping 192.168.50.1 # EnterpriseLAN - SUCCESS  
C:\> ping 192.168.60.1 # Monitoring - SUCCESS
```

Test 2: Service Access Verification - pfSense Web GUI: `https://192.168.10.1`  Accessible - **Wazuh SIEM Dashboard:** `https://192.168.20.2`  Accessible

- **Grafana Monitoring:** `http://192.168.60.2:3000`  Accessible - **Prometheus Metrics:** `http://192.168.60.2:9090`  Accessible

Test 3: SSH Connectivity - Ansible Controller: `ssh nantwi@192.168.10.2`  Success - **Wazuh Server:** `ssh nantwi@192.168.20.2`  Success - **Monitoring Server:** `ssh nantwi@192.168.60.2`  Success

Final Network Configuration

Tailscale Device Inventory

Device	Tailscale IP	Role	Physical Location	Status
pfsense-homelab	100.94.120.6	Subnet Router	Lab Infrastructure	 Connected
nantwi	100.90.17.43	Admin Client	Windows Desktop	 Connected

Advertised Subnet Routes

VLAN	Subnet	Gateway	Status	Services
Management	192.168.10.0/24	.1	 Active	pfSense GUI, Ansible Controller
BlueTeam	192.168.20.0/24	.1	 Active	Wazuh SIEM (All-in-One)
RedTeam	192.168.30.0/24	.1	 Active	Reserved for Penetration Testing
DevOps	192.168.40.0/24	.1	 Active	Reserved for CI/CD Pipeline
EnterpriseLAN	192.168.50.0/24	.1	 Active	Reserved for Business Services
Monitoring	192.168.60.0/24	.1	 Active	Grafana, Prometheus

Security Implementation

Encryption and Protocol

- **Protocol:** WireGuard with ChaCha20Poly1305 encryption
- **Key Management:** Automatic rotation every 180 days
- **Authentication:** Google OAuth integration
- **Device Authorization:** Manual approval required

Network Security Maintained

- **VLAN Isolation:** Preserved through pfSense firewall rules
- **Inter-VLAN Communication:** Controlled by existing pfSense policies
- **Zero Trust:** No implicit network access assumptions

Results Achieved

Operational Benefits

- ✓ **Bypassed ISP Restrictions:** No upstream router configuration required
- ✓ **Secure Remote Access:** All lab VLANs accessible worldwide
- ✓ **Zero Client Configuration:** Install app, authenticate, connect
- ✓ **Modern Security:** WireGuard encryption with automatic key rotation
- ✓ **Scalable Architecture:** 98 additional devices supported
- ✓ **Universal Compatibility:** Works from any network environment

Performance Metrics

- **Connection Latency:** Minimal overhead (< 10ms additional)
- **Throughput:** No significant bandwidth limitations observed
- **Reliability:** Automatic DERP relay failover
- **Device Capacity:** 2/100 devices utilized (2% of free plan)

Comparison: Tailscale vs. Original OpenVPN Plan

Aspect	OpenVPN (Planned)	Tailscale (Implemented)
ISP Requirements	✗ Port forwarding mandatory	✓ No ISP interaction needed
Setup Time	✗ Hours (certificates, configs)	✓ Minutes (OAuth + install)
Client Setup	✗ Manual config files	✓ Zero configuration
VLAN Support	✗ Complex routing setup	✓ Single command advertisement
Maintenance	✗ Certificate renewals	✓ Fully managed service
Security	✗ Manual key management	✓ Automatic key rotation
Reliability	✗ Single point of failure	✓ Distributed mesh network
Cross-Platform	✗ Per-device configuration	✓ Native apps all platforms

Troubleshooting Notes

FreeBSD Service Issues

Problem: Standard service commands failed on pfSense

```
service tailscaled start
# Error: Cannot 'start' tailscaled
```

Solution: Manual daemon initialization

```
mkdir -p /var/lib/tailscale
/usr/local/bin/tailscaled --state=/var/lib/tailscale/tailscaled.state --socket=/var/run/tailscale/
```

Package Location Discovery

Issue: `tailscaled` not found in PATH after installation

Resolution: Located binaries in `/usr/local/bin/` - `/usr/local/bin/tailscale` (client) - `/usr/local/bin/tailscaled` (daemon)

Success Metrics

Implementation Success Criteria - ALL ACHIEVED

-  **Remote Access Established** - All 6 VLANs accessible externally
 -  **ISP Independence** - Zero upstream router dependencies
 -  **Service Accessibility** - pfSense, Wazuh, Grafana, Prometheus reachable
 -  **Security Maintained** - Modern encryption with zero trust principles
 -  **Operational Efficiency** - Single-click connection for users
 -  **Scalability Ready** - Architecture supports 98 additional devices
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Conclusion

The Tailscale implementation successfully transformed the homelab from a locally-accessible infrastructure to a globally-accessible, enterprise-grade environment. By bypassing ISP router limitations through modern mesh networking, the solution provides superior security, performance, and management capabilities compared to traditional VPN approaches.

Key Achievement: Converted a significant network limitation (no ISP router access) into a competitive advantage by implementing a more modern, secure, and scalable remote access solution than originally planned with OpenVPN.

The implementation enables secure remote management of the entire lab infrastructure - including Blue Team security tools (Wazuh SIEM), monitoring systems (Grafana/Prometheus), and administrative interfaces (pfSense) - accessible from any location worldwide with enterprise-grade WireGuard encryption.

Implementation Date: July 2025 | Status: Production Ready | Next Phase: Lab Service Expansion